

LECTURE 7: SQUARE MATRICES

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PLAN

Square matrices

1. Identity, diagonal and symmetric matrices
2. Gaussian elimination
3. Inverse
4. Trace and determinant



SQUARE MATRICES

Consider $\mathbb{R}^{n \times n}$, the set of n by n **square** matrices.

If $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ then $\mathbf{A} + \mathbf{B}$, $c\mathbf{A}$ with $c \in \mathbb{R}$ and $\mathbf{AB}$ are also in $\mathbb{R}^{n \times n}$. Also $\mathbf{A}^\top \in \mathbb{R}^{n \times n}$.

We define the **identity matrix** $\mathbf{I}_n \in \mathbb{R}^{n \times n}$ as the matrix

$$\mathbf{I}_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

We have $\mathbf{I}_n \mathbf{A} = \mathbf{A} \mathbf{I}_n = \mathbf{A}$ for every $\mathbf{A} \in \mathbb{R}^{n \times n}$.

DIAGONAL AND TRIANGULAR MATRICES

Given $a_1, \dots, a_n \in \mathbb{R}$ we define the **diagonal matrix**

$$\text{diag}(a_1, \dots, a_n) = \begin{pmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{pmatrix}.$$

We say that $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **lower/upper triangular** if it has zeros above/below the diagonal:

$$\begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

lower triangular

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}$$

upper triangular

SYMMETRIC MATRICES

We say that $\mathbf{A}$ is **symmetric** if $\mathbf{A}^\top = \mathbf{A}$.

In other words, if the rows of $\mathbf{A}$ equal the columns of $\mathbf{A}$.

Examples.

- $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}$ is symmetric.
- $\begin{pmatrix} 1 & 2 & \mathbf{3} \\ 2 & 4 & 5 \\ \mathbf{7} & 5 & 6 \end{pmatrix}$ is not.

QUESTION 1

Convince yourself that if $\mathbf{A} \in \mathbb{R}^{m \times n}$ then $\mathbf{A}^\top \mathbf{A}$ is symmetric.

GAUSSIAN ELIMINATION

We want to solve a system of linear equations $A\mathbf{x} = \mathbf{b}$.

Idea: transform A into an upper triangular matrix using transformations that are equivalent to left-multiplying by matrices C . If we do the same to $\mathbf{b}$, we preserve the equation: $A\mathbf{x} = \mathbf{b}$ implies $CA\mathbf{x} = C\mathbf{b}$.

We can:

- Take one row and multiply it by a nonzero scalar.
- Take one row and add it to another one, possibly multiplied by some scalar.
- Exchange two rows.

Once A is upper triangular, the system of equations is easy.

GAUSSIAN ELIMINATION

We want to solve

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Let's clean the first column. Take the first row and subtract it from the second one. Let's keep track of the changes, ignoring $\mathbf{x}$ for now:

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & -2 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{array} \right).$$

Let's clean the second column. We want to use the second row, so let's first divide it by -2 :

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 1 & 1 & 1 \end{array} \right).$$

Now subtract it from the third row.

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & \frac{3}{2} & 1 \end{array} \right).$$

Great! Now the equation is

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & \frac{3}{2} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \text{i.e.,} \quad \begin{cases} x_1 + x_2 = 1 \\ x_2 - \frac{1}{2}x_3 = 0 \\ \frac{3}{2}x_3 = 1 \end{cases}$$

so $x_3 = \frac{2}{3}$, $x_2 = \frac{1}{2}x_3 = \frac{1}{3}$, and $x_1 = 1 - x_2 = \frac{1}{3}$.

Example. We want to solve
$$\begin{cases} x_1 + x_2 + x_3 = 1 \\ x_1 + x_2 - x_3 = 0 \\ x_2 + x_3 = 1 \end{cases}.$$

Gauss Algorithm:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 1 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & -2 & -1 \\ 0 & 1 & 1 & 1 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & -2 & -1 \end{array} \right)$$

Notice. We had to exchange rows in the second step.

The system becomes

$$\begin{cases} x_1 + x_2 + x_3 = 1 \\ x_2 + x_3 = 1 \\ -2x_3 = -1 \end{cases},$$

so $x_3 = \frac{1}{2}$, $x_2 = \frac{1}{2}$, and $x_1 = 1 - x_2 - x_3 = 0$.

EXERCISE

QUESTION 2

Solve

$$\begin{cases} x_1 + x_2 + x_3 = 1 \\ x_1 + x_2 - x_3 = 0 \\ 2x_1 \quad \quad + x_3 = 1. \end{cases}$$

using Gaussian elimination.

INVERSE

Given $\mathbf{A} \in \mathbb{R}^{n \times n}$ we say that $\mathbf{A}^{-1} \in \mathbb{R}^{n \times n}$ is the **inverse** of $\mathbf{A}$ if

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_n.$$

Not every nonzero matrix has an inverse. For example, suppose that

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

has an inverse $\mathbf{A}^{-1}$. We have $\mathbf{A} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1-1 \\ 1-1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \mathbf{0}.$

Multiplying by $\mathbf{A}^{-1}$ the LHS we get: $\mathbf{A}^{-1}\mathbf{A} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \mathbf{I}_n \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$ But multiplying the RHS we get $\mathbf{A}^{-1}\mathbf{0} = \mathbf{0} \neq \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$ This is impossible. Therefore, $\mathbf{A}^{-1}$ cannot exist.

PROPERTIES OF THE INVERSE

If $\mathbf{A}$ has an inverse we say that $\mathbf{A}$ is **invertible**.

Suppose that $\mathbf{A}, \mathbf{B}$ are invertible. Then

- The inverse is unique.
- $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$.
- $(c\mathbf{A})^{-1} = c^{-1}\mathbf{A}^{-1}$ if $c \in \mathbb{R}, c \neq 0$.
- $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.
- $(\mathbf{A}^{-1})^{\top} = (\mathbf{A}^{\top})^{-1}$.
- $\text{diag}(a_1, \dots, a_n)^{-1} = \text{diag}(a_1^{-1}, \dots, a_n^{-1})$ if $a_1, \dots, a_n \in \mathbb{R} \setminus \{0\}$.

QUESTION 3

Why?

CALCULATING THE INVERSE

Given $\mathbf{A} \in \mathbb{R}^{n \times n}$ we want $\mathbf{A}^{-1}$ such that $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_n$.

We can use Gauss Algorithm: if we do transformations to $\mathbf{A}$ that are left-multiplying by $\mathbf{C}$, and we end up with $\mathbf{C}\mathbf{A} = \mathbf{I}_n$, then $\mathbf{A}^{-1} = \mathbf{C} = \mathbf{C}\mathbf{I}_n$, so $\mathbf{A}^{-1}$ is the same as doing those transformations to $\mathbf{I}_n$.

Example. We want $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1}$.

$$\left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{array} \right) \Rightarrow \left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & -2 & -1 & 1 \end{array} \right) \Rightarrow \left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{1}{2} & -\frac{1}{2} \end{array} \right) \Rightarrow \left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} & -\frac{1}{2} \end{array} \right)$$

$$\text{So } \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}.$$

Example. Let's try to find $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}^{-1}$.

$$\left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right) \Rightarrow \left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{array} \right) \Rightarrow ?$$

If we get an upper triangular matrix with a zero in the diagonal, we won't be able to invert it.

EXERCISE

QUESTION 4

Calculate $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix}^{-1}$ if it exists.

QUESTION 5

Calculate $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}^{-1}$ if it exists.

EXERCISE

QUESTION 6

Verify that $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ if $ad - bc \neq 0$.

What happens if $ad - bc = 0$?

TRACE

The **trace** of a square matrix is the sum of its diagonal elements. It defines the function $\text{tr} : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ given by

$$\text{tr}(\mathbf{A}) = a_{11} + \cdots + a_{nn}.$$

Example. $\text{tr} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -1 \\ -1 & 1 & -1 \end{pmatrix} = 1 + 2 - 1 = 2.$

Properties.

- $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B}),$
- $\text{tr}(c\mathbf{A}) = c \text{tr}(\mathbf{A})$ if $c \in \mathbb{R},$
- $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ even if $\mathbf{A}, \mathbf{B}$ are not square, as long as $\mathbf{AB}$ and $\mathbf{BA}$ are,
- $\text{tr}(\mathbf{A}^\top) = \text{tr}(\mathbf{A}).$

EXERCISE

QUESTION 7

Convince yourself that $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B})$ and $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$.

DETERMINANT

The **determinant** of a square matrix A , $\det(\mathbf{A})$, is defined recursively as follows:

If $\mathbf{a} \in \mathbb{R}^{1 \times 1}$ then $\det(\mathbf{a}) = \mathbf{a}$.

Given

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix},$$

we take the first column, and for each cell a_{i1} we delete the row and column of that cell, obtaining a matrix $\mathbf{M}_{i1}$. The determinant is then

$$\det(\mathbf{A}) = a_{11} \det(\mathbf{M}_{11}) - a_{21} \det(\mathbf{M}_{21}) + \cdots + (-1)^{n+1} a_{n1} \det(\mathbf{M}_{n1}).$$

We can write $|\mathbf{A}|$ instead of $\det(\mathbf{A})$.

EXAMPLE

$$\det \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -1 \\ -1 & 1 & -1 \end{pmatrix} = 1 \cdot \det \begin{pmatrix} 2 & -1 \\ 1 & -1 \end{pmatrix} - 1 \cdot \det \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} + (-1) \cdot \det \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}$$

Now,

- $\det \begin{pmatrix} 2 & -1 \\ 1 & -1 \end{pmatrix} = 2 \cdot (-1) - 1 \cdot (-1) = -2 + 1 = -1,$
- $\det \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = 1 \cdot (-1) - 1 \cdot 1 = -2,$ and
- $\det \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} = 1 \cdot (-1) - 2 \cdot 1 = -1 - 2 = -3.$

$$\text{Therefore, } \det \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -1 \\ -1 & 1 & -1 \end{pmatrix} = 1 \cdot (-1) - 1 \cdot (-2) + (-1) \cdot (-3) = -1 + 2 + 3 = 4.$$

DIAGONAL AND TRIANGULAR MATRICES

Examples.

- $\det \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = 2 \cdot \det \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = 2 \cdot 1 \cdot \det(-1) = 2 \cdot 1 \cdot (-1) = -2.$

- $\det \begin{pmatrix} 2 & 5 & 3 \\ 0 & 1 & 7 \\ 0 & 0 & -1 \end{pmatrix} = 2 \cdot \det \begin{pmatrix} 1 & 7 \\ 0 & -1 \end{pmatrix} = 2 \cdot 1 \cdot \det(-1) = 2 \cdot 1 \cdot (-1) = -2.$

In general, if $\mathbf{A}$ is diagonal or triangular then $\det(\mathbf{A})$ is the product of the numbers in the diagonal.

EXERCISE

QUESTION 8

Calculate $\det \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix}$.

PROPERTIES

If $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ and $c \in \mathbb{R}$ then

- $\det(c\mathbf{A}) = c^n \det(\mathbf{A})$,
- $\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B})$,
- $\det(\mathbf{I}_n) = 1$,
- $\mathbf{A}$ is invertible iff $\det(\mathbf{A}) \neq 0$; in that case, $\det(\mathbf{A}^{-1}) = \det(\mathbf{A})^{-1}$,
- $\det(\mathbf{A}^\top) = \det(\mathbf{A})$.

ADJOINT MATRIX

Given $\mathbf{A} \in \mathbb{R}^{n \times n}$ let M_{ij} be the determinant of the matrix that results from removing the i th row and the j th column of $\mathbf{A}$.

The **adjoint** of $\mathbf{A}$ is the n by n matrix $\mathbf{adj}(\mathbf{A})$ whose ij entry is $(-1)^{i+j}M_{ji}$.

Example.

$$\mathbf{adj} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -1 \\ -1 & 1 & -1 \end{pmatrix} = \begin{pmatrix} \det \begin{pmatrix} 2 & -1 \\ 1 & -1 \end{pmatrix} & -\det \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix} & \det \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} \\ -\det \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} & \det \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} & -\det \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \\ \det \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} & -\det \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} & \det \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} \end{pmatrix}^{\top} = \begin{pmatrix} -1 & 2 & 3 \\ 2 & 0 & -2 \\ -3 & 2 & 1 \end{pmatrix}^{\top}$$

Key Property. $\mathbf{adj}(\mathbf{A})\mathbf{A} = \det(\mathbf{A})\mathbf{I}_n$, so $\mathbf{A}^{-1} = \det(\mathbf{A})^{-1}\mathbf{adj}(\mathbf{A})$.

EXERCISE

QUESTION 9

Use the formula $\mathbf{A}^{-1} = \det(\mathbf{A})^{-1}\text{adj}(\mathbf{A})$ to calculate $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix}^{-1}$.

